

ITDSA2-12 Project 2

Site: [Eduvos LMS](#)
Course: Database Systems Assessments
Book: ITDSA2-12 Project 2

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1. ITDSA2-12 Project 2

Faculty Name:	Information Technology
Module Name:	Database Systems
Module Code:	ITDSA2-12
Date:	Week 6

Total Marks: 100

Examiner: Obaro Adewale

NOTE

5% will be deducted from the student's assignment mark for each calendar day the assignment is submitted late, up to a maximum of three calendar days. The penalty will be based on the official campus submission date.

Projects submitted later than three calendar days after the deadline or not submitted will get 0%.¹ This is an **individual** project.

Question	Marks Subtotal
Question 1: Scenario-Based	50 Marks
Question 2: Scenario-Based	50 Marks
Question 3: Application-Based	50 Marks

IMPORTANT NOTICE:

1. ATTEMPT ANY TWO QUESTION/DELIVERABLE ONLY

By attempting this assessment you confirm that you have read and understood the Examination Rules.

2. AI Checklist and Declaration

Before you submit an assignment, you should be able to confidently and honestly make all the below statements. For group work, you can also review the list, together, to hold one another accountable.

- I confirm that my submission reflects my personal learning, knowledge, skills, and understanding.
- If AI tools were employed for generating any part of this assignment (even in the drafting/research phase), I have referenced the use of AI in the text and/or declared the use of AI. I am willing to discuss the process and its contribution to my learning.
- I am aware that the lecturer may request a demonstration of my learning, such as explaining choices in approach, research, and the content I am submitting.
- I am aware that, if I did use AI in any phase of preparing this submitted work, it is recommended that I save a copy of the relevant chat history (prompts and answers), as this will help me demonstrate my writing/work process to my lecturer, if I am asked to do so.
- I have read the assignment instructions on whether AI tools are prohibited for this assignment, and if they are prohibited, I can confirm that I did not use AI tools.

- I understand that failure to agree to these terms may be deemed unethical, potentially leading to disciplinary action. I understand my responsibility for the integrity of my work, including seeking clarification from academic staff and adhering to instructions.

It is essential to acknowledge your use of ChatGPT or other generative AI in your learning. If you use ChatGPT or other generative AI to help you generate ideas or plan your process, you should still acknowledge how you used the tool, even if you don't include any AI-generated content in the assignment.

Please note: The following guiding questions that you will be asked in an AI declaration questionnaire below this assignment brief.

AI Declaration

It is compulsory to complete this AI declaration for each of your assignment submissions.

I carefully read the assignment instructions, and the extent to which AI may be used for the assignment.
I used the following AI system(s)/tool(s):
I used it for the following:
If I quoted or paraphrased an AI output, I have referenced the relevant tool, version, and the date I used the tool.
I still consider this work my own. (i.e., I have not outsourced the final product, or significant portions of it, to AI tools/systems).
If required, I can defend my argument/perspective, explain my choices and approach, and can show that I am knowledgeable about the details of my work.

For further guidance on the use of AI at Eduvos, please refer to the AI FAQ glossary. You will locate the FAQs in the Artificial Intelligence tile on the myDocuments page of myLMS.

3. Instructions to Students

1. Please ensure that your answer file (where applicable) is named as follows before submission: **Module Code – Assessment Type – Campus Name – Student Number.**
2. Remember to keep a copy of all submitted assignments.
3. All work must be typed.
4. Please note that you will be evaluated on your writing skills in all your assignments.
5. All work must be submitted through Turnitin. The full originality report will be automatically generated and available for the lecturer to assess. Negative marking will be applied if you are found guilty of plagiarism, poor writing skills, or if you have applied incorrect or insufficient referencing. (See the "instructions to students" book activity before this activity where the application of negative marking is explained.)
6. You are not allowed to offer your work for sale or to purchase the work of other students. This includes the use of professional assignment writers and websites, such as Essay Box. You are also not allowed to make use of artificial

intelligence tools, such as ChatGPT, to create content and submit it as your own work. If this should happen, Eduvos reserves the right not to accept future submissions from you.

4. Special Considerations

Learning Objective

This project is designed to evaluate the fundamental and functional understanding of relational and Graph database design.

Scope

Scope of each question is provided at the question level of the project.

Technical Requirements

The technical requirements of this project are as follows:

1. A computer with (Compulsory):

- Oracle DBMS Software
- Mongo DBMS
- Neo4j DBMS

2. Suitable Online DBMS Platforms for Database Systems CRUD operations

- Online Oracle platform 1: <https://livesql.oracle.com/apex/f?p=590:1000>
- Online Oracle platform 2: <https://livesql.oracle.com/next/>
- MongoDB Online IDE: <https://www.mycompiler.io/new/mongodb>
- Neo4j Online Compiler: <https://console.neo4j.org/>

3. Students should **ALWAYS** save their work progress in case of unforeseen circumstance(s).

4. All queries or scripts **MUST BE TYPED** into the text area provided or into a word document and uploaded as a single document.

5. Screenshot(s) of text or typed text is **NOT ALLOWED**. Screenshot of any typed codes, queries or scripts will automatically be awarded **ZERO**.

6. All text must be typed into a word document, which is to be uploaded through an Assignment / Turnintin Submission link.

7. Submit the files with code, data set, and a Word document explaining the problem and queries you have created with a screenshot of the result.

Marking Criteria

Please refer to the rubric for the mark allocations of the project.

Kindly **award zero for any screenshots of texts** as all texts must be typed and screenshots are not acceptable. This was duly communicated to the students in the technical requirements of this project.

Submission Dates and Mark Allocation of Deliverables

Please note that each student is expected to individually attempt **ANY TWO (2) deliverables** of their choice.

1. Deliverable 1 Mark Allocation (**Database Implementation Using Oracle DBMS**): 50
2. Deliverable 2 Mark Allocation (**Database Implementation Using Mongo DBMS**): 50
3. Deliverable 3: Mark Allocation (**Database Implementation Using NEO4J DBMS**): 50

Submission Week: **Week 8**

Final Mark Calculation:

Final Mark: (Deliverable 1 Mark + Deliverable 2 Mark)

OR

Final Mark: (Deliverable 1 Mark + Deliverable 3 Mark)

OR

Final Mark: (Deliverable 2 Mark + Deliverable 3 Mark)

5. Question 1

**Deliverable 1: Database
system using Oracle DBMS**

50 Marks

Study the scenario and complete the question(s) that follow:

Modise Cultural Facility Management System

Modise cultural facility is a multi-purpose venue designed to host cultural activities, such as performances, exhibitions, workshops, and community gatherings. It promotes artistic expression, cultural diversity, and community engagement.

Facilities designated for cultural activities are identified by a unique identity number and are characterized by a name, capacity, and a specific address within a municipality. Each facility may contain multiple rooms, with each room represented by a number accompanied by a description, such as theatre, cinema.

A municipality is described by an official code, a name and it is located within in a province. The average number of people living in each municipality are also kept for demographic and socio-development purposes. Since provinces may encompass multiple municipalities, it is essential to specify the province to which each municipality belongs. The provinces are identified by a unique code and name, which are used to associate relevant municipalities with their respective provinces.

Various activities which include but not limited to music concert, IsiZulu dance competition and art exhibition take place in the cultural facilities situated within a province. The activity categories may include but not limited to cinema, circus, dance, music, bullfights and theatre. Since a single facility can host activities from multiple categories, each entry is captured in an intermediate entity using the facility's ID and the reference of an activity taking place there. In addition, the date on which an activity occur is also stored. An Entity Relationship Diagram (ERD) for the scenario is presented in the Figure 1.

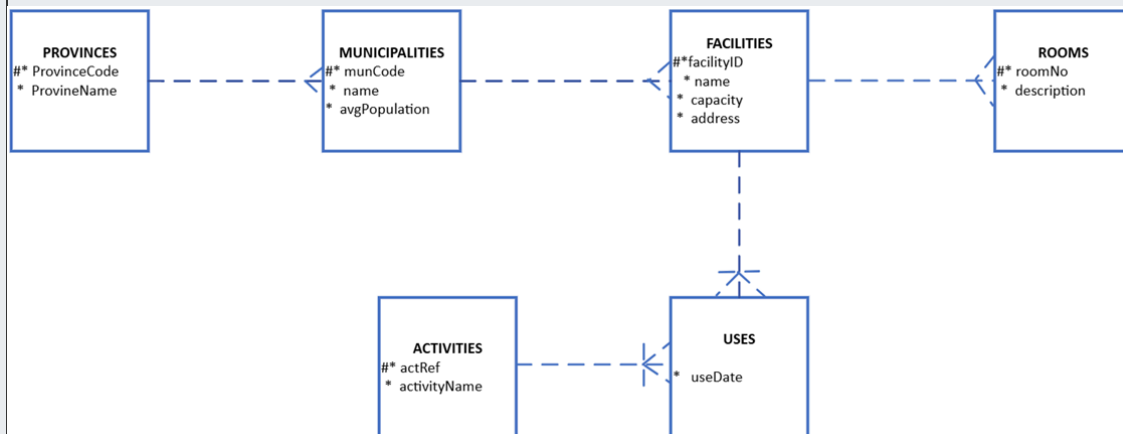


Figure 1: ERD for the Scenario presented in deliverable 1.

Source: Obaro Adewale (2025)

Note:

- All questions must be answered using queries only. All queries must be typed in formatted text. As such, screenshot(s)/images of queries is completely unacceptable and would attract zero.
- All queries in this section must be written in Oracle DBMS platform only.
- Queries written using another database IDE apart from Oracle platform would automatically attract zero.

Questions for Consideration Under Deliverable 1

Use the above scenario to answer the following questions:

1.1 Create the tables identified in the ERD from the given scenario.

(18 Marks)

1.2 Create a UNIQUE sequence which increment by one (1) for the tables created in Question 1.1. Therefore, no sequence should have the same value, and it is strongly advised NOT to create a sequence for USES entity because it is an intersection entity.

(5 Marks)

1.3 Store at least two (2) records into each table created in Question 1.1 using INSERT statement and relevant sample data. Use the sequence created for each table in Question 1.2 to automatically insert values in the primary key column. It is important to note that sequence is not expected to be implemented for USES entity because it is an intersection entity without a specific primary key column as evidenced in the ERD. Therefore, a composite key comprising of the two foreign keys from the parent tables should rather be implemented in USES table.

(12 Marks)

1.4 Retrieve and display the number of municipalities that do not have any facility with an activity of 'Music' using a suitable query. Remember to use multiple JOIN statements, aggregate functions, suitable wild cards character and sub-query where necessary.

(4 Marks)

1.5 Write a query to retrieve and display a list of provinces that have municipalities with an average population of 4,000,000 (4 million) or more. You may consider the use of sub-query and relevant conditional operators in obtaining the output if necessary.

(3 Marks)

1.6 The management intends to know the utilisation capacity of each facility in order to plan for future events and expansion. Write a stored procedure named **"Province_Capacity_Utilization_procedure"** that calculates and displays the cultural facility capacity utilisation per province. The procedure should:

- Calculates the total number of facilities in each province.

(1 Mark)

- Sum up the total capacity of all facilities within each province.

(1 Mark)

- Count the total number of activities hosted by those facilities.

(1 Mark)

- Computes a capacity utilisation percentage using the formula below:

$$Utilization (\%) = \left[\frac{Total\ Activities}{Total\ Capacity} \right] \times 100$$

(2 Marks)

- Display a province-based report that includes the province name, number of facilities, total capacity, total activities hosted and the computed utilisation percentage in each province.

(2 Marks)

- Sort the output by utilisation percentage in descending order.

(1 Mark)

```
Statement processed.  
Province: Gauteng | Facilities: 1 | Capacity: 300 | Activities: 1 | Utilization: .33%  
Province: Western Cape | Facilities: 1 | Capacity: 500 | Activities: 1 | Utilization: .2%
```

(8 Marks)

End of Question 1

6. Question 2

Deliverable 2: Implementation of Database System with MongoDB

50 Marks

Study the scenario 1 and complete the question(s) that follow:

MongoDB Implementation: Modise Cultural Facility Management System

Implement the Modise Cultural Facility scenario using MongoDB. Design an appropriate data model that captures the hierarchical relationships among provinces, municipalities, facilities, rooms, and activities. You may choose between embedding related documents or referencing them across collections. Ensure that your implementation supports efficient querying for activities by date, facility details, and their associated municipalities and provinces. Provide sample documents and queries that demonstrate the functionality of your data model. The following sample data may be considered in the database implementation. The following sample data may be considered for implementation of the database system.

Province	
provinceCode	name
WC	Western Cape
KZN	KwaZulu-Natal

Municipality			
munCode	name	avgPopulation	provinceCode
CPT	City of Cape Town	4005016	WC
ETH	eThekweni Metropolitan Municipality	3702231	KZN

Facility				
facilityId	name	address	capacity	munCode
Arts	Artscape Theatre Centre	D.F. Malan St, Foreshore, Cape Town, 8001	1500	CPT
ICC	Durban ICC	45 Bram Fischer Rd, Durban Central, Durban, 4001	10000	ETH

Room		
roomNo	description	facilityId
1	Hall 1	Arts
2	Hall 2	ICC

Activity

activityRef	name
MUS01	Music Concert
EXH01	Art Exhibition

USES		
facilityId	activityRef	Date
Arts	MUS01	2025-05-20
ICC	EXH01	2025-06-15

Sample data for consideration.

Source: Obaro Adewale (2025)

Note:

- All questions must be answered using queries only. All queries must be typed in formatted text. As such, screenshot(s)/images of queries is completely unacceptable and would attract zero.
- All queries in this section must be written in MongoDB DBMS platform only.
- Queries written using another database IDE apart from MongoDB platform would automatically attract zero.

Questions for Consideration Under Deliverable 2

Use the above scenario to answer the following questions:

2.1 Create a database called **OLMS** using query.

(1 Mark)

2.2 Create the collections identified / derived from the ERD in the given scenario.

(5 Marks)

2.3 Store at least **two (2)** records into each collection in Question 2.2.

(18 Marks)

2.4 Write a query to retrieve and display the name of a facility such as "Arts" and all the related rooms associated with the facility. The result of the query may be displayed is follow:

```
[
  {
    _id: ObjectId('67fbe0f668de0cf97f3c71f2'),
    facilityId: 'Arts',
    name: 'Artscape Theatre Centre',
    capacity: 1500,
    address: 'D.F. Malan St, Foreshore, Cape Town, 8001',
    munCode: 'CPT',
    rooms: [
      { roomNo: 1, description: 'Main Theatre' },
      { roomNo: 2, description: 'Opera House' }
    ]
  }
]
```

(3 Marks)

2.5 Write a query to retrieve and display a list of municipalities with an average population of 3,500,000 (3.5 million) or more. The result below demonstrates the result of the query.

```
[
  {
    _id: 'CPT',
    name: 'City of Cape Town',
    avgPopulation: 4005016,
    provinceCode: 'WC'
  },
  {
    _id: 'ETH',
    name: 'eThekweni Metropolitan Municipality',
    avgPopulation: 3702231,
    provinceCode: 'KZN'
  }
]
```

(2 Marks)

2.6 Retrieve and display the date on which a facility is used, the details of the facility including facility identity, name, capacity and address. A sample output of the query result is displayed as follow:

```
[
  {
    useDate: ISODate('2025-05-20T00:00:00.000Z'),
    facilityId: 'Arts',
    facilityName: 'Artscape Theatre Centre',
    capacity: 1500,
    address: 'D.F. Malan St, Foreshore, Cape Town, 8001'
  },
  {
    useDate: ISODate('2025-06-15T00:00:00.000Z'),
    facilityId: 'ICC',
    facilityName: 'Durban ICC',
    capacity: 10000,
    address: '45 Bram Fischer Rd, Durban Central, Durban, 4001'
  }
]
ModiseDB> |
```

(5 Marks)

2.7 Write a query to retrieve and display a list of activities including activity reference codes, activity name along with the date on which each activity takes place as well as detailed information regarding the facility such as the facility identity, facility name, capacity and address of each facility. In addition, the municipalities and province in which the facility are situated should be displayed with other requested information. This task requires obtaining information from multiple collections, with a sample output of the query as shown below:

```
[
  {
    useDate: ISODate('2025-05-20T00:00:00.000Z'),
    activityRef: 'MUS01',
    activityName: 'Music Concert',
    facilityId: 'Arts',
    facilityName: 'Artscape Theatre Centre',
    capacity: 1500,
    address: 'D.F. Malan St, Foreshore, Cape Town, 8001',
    municipality: 'City of Cape Town',
    province: 'Western Cape'
  },
  {
    useDate: ISODate('2025-06-15T00:00:00.000Z'),
    activityRef: 'EXH01',
    activityName: 'Art Exhibition',
    facilityId: 'ICC',
    facilityName: 'Durban ICC',
    capacity: 10000,
    address: '45 Bram Fischer Rd, Durban Central, Durban, 4001',
    municipality: 'eThekweni Metropolitan Municipality',
    province: 'KwaZulu-Natal'
  }
]
```

(16 Marks)

End of Question 2

7. Question 3

Deliverable 3: Implementation of Database System with Neo4j DBMS

50 Marks

Study the scenario 1 and complete the question(s) that follow:

Neo4j Implementation: Modise Cultural Facility Management System

Implement the Modise Cultural Facility scenario using Neo4j dbms that accurately represents entities such as provinces, municipalities, facilities, rooms, and activities as nodes, and their interconnections as relationships. Ensure that your model leverages Neo4j's strengths by using specific relationship types and appropriate directionality to reflect the nature of the connections. Incorporate relevant properties for both nodes and relationships to capture essential details, such as activity dates or facility capacities. Provide sample Cypher queries that demonstrate the retrieval of complex information, such as activities hosted by facilities within specific municipalities and provinces on given dates. The following sample data may be considered for implementation of the database system.

Province	
provinceCode	name
GP	Gauteng
WC	Western Cape

Municipality			
munCode	name	averagePopulation	provinceCode
JHB	City of Johannesburg	4949347	GP
CPT	City of Cape Town	4005016	WC

Facility				
facilityId	name	address	capacity	munCode
FAC001	Modise Cultural Facility	123 Cultural St	500	JHB
FAC002	Ubuntu Arts Centre	456 Ubuntu Ave	300	CPT

Room		
roomNo	description	facilityId
R001	Theatre	FAC001
R002	Cinema	FAC001
R003	Exhibition Hall	FAC002
R004	Workshop Room	FAC002

Activity	
activityRef	name

ACT001	IsiZulu Dance Competition
ACT002	Art Exhibition
ACT003	Music Concert
ACT004	Theatre Performance

USES		
facilityId	activityRef	Date
FAC001	ACT001	2025-05-20
FAC001	ACT002	2025-06-20
FAC002	ACT003	2025-10-20
FAC002	ACT004	2025-12-12

Sample data for consideration.

Source: Obaro Adewale (2025)

Note:

- All questions must be answered using queries only. All queries must be typed in formatted text. As such, screenshot(s)/images of queries is completely unacceptable and would attract zero.
- All queries in this section must be written in Neo4j DBMS platform only.
- Queries written using another database IDE apart from Neo4j platform would automatically attract zero.

Questions for Consideration Under Deliverable 3

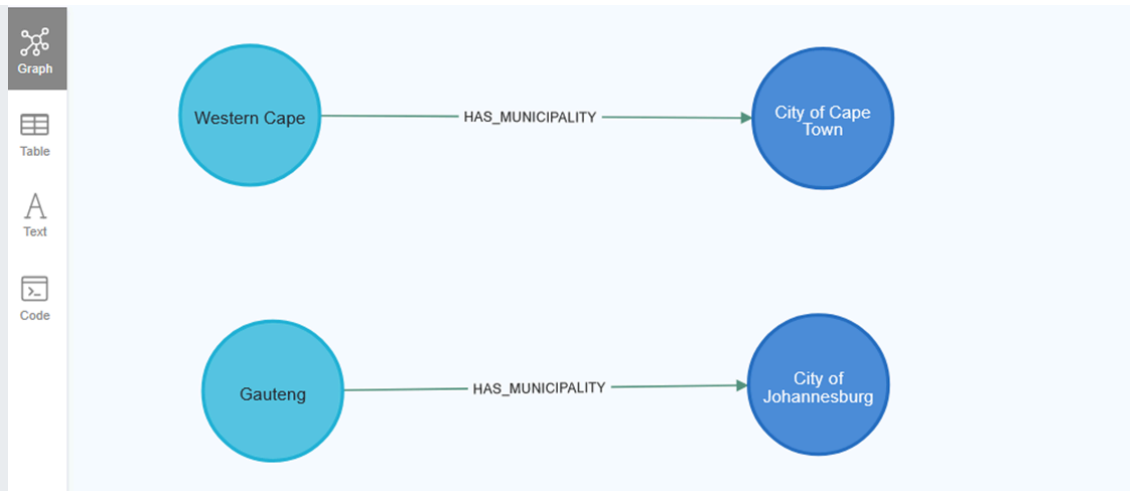
Use the information provided to answer the following questions:

3.1 Create a database called ONLMS **using query**.

(2 Marks)

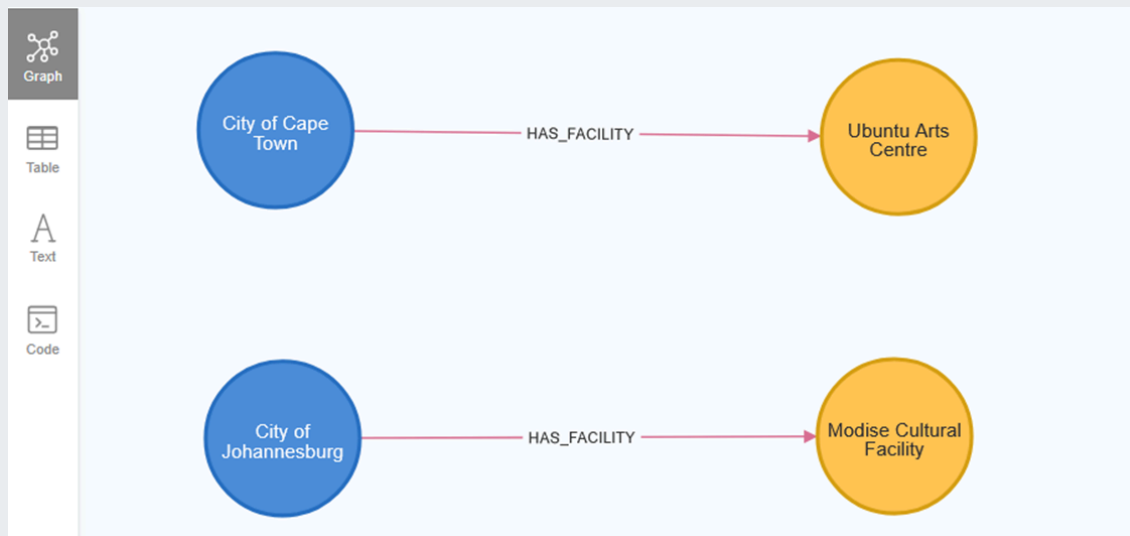
3.2 Using the provided sample data or sample data of choice, store at least two (2) data nodes and establish the relationships that exist between the collections (tables):

(a) Data nodes of province and municipality as well as related relationship. The output of the query results may be displayed in the format below:



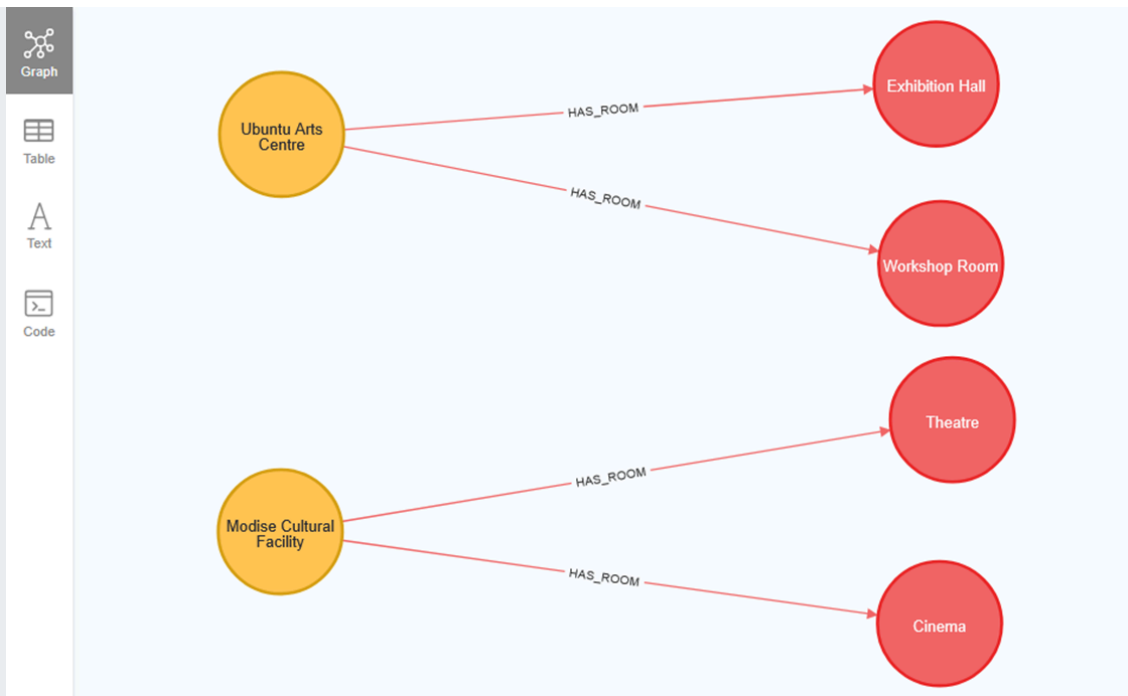
(6 marks)

(b) Data nodes of municipality and facility as well as related relationship. The output of the query results may be displayed in the format below:



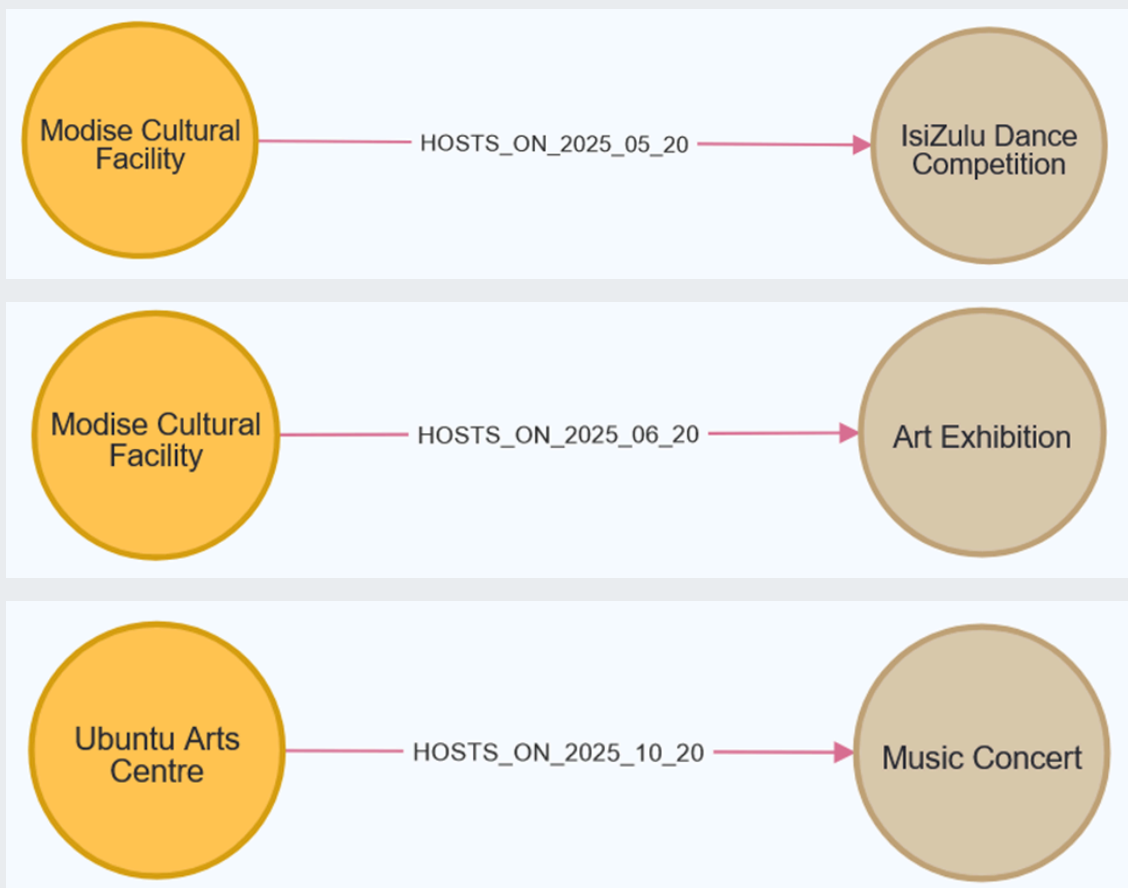
(4 marks)

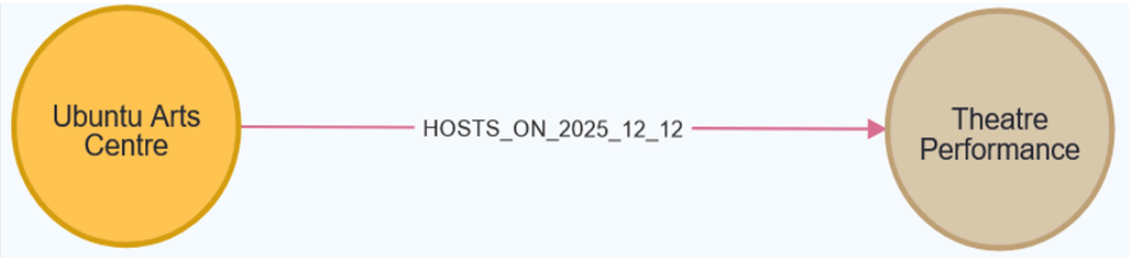
(c) Data nodes of facility and room as well as associated relationship. The output of the query results may be displayed in the format below:



(8 marks)

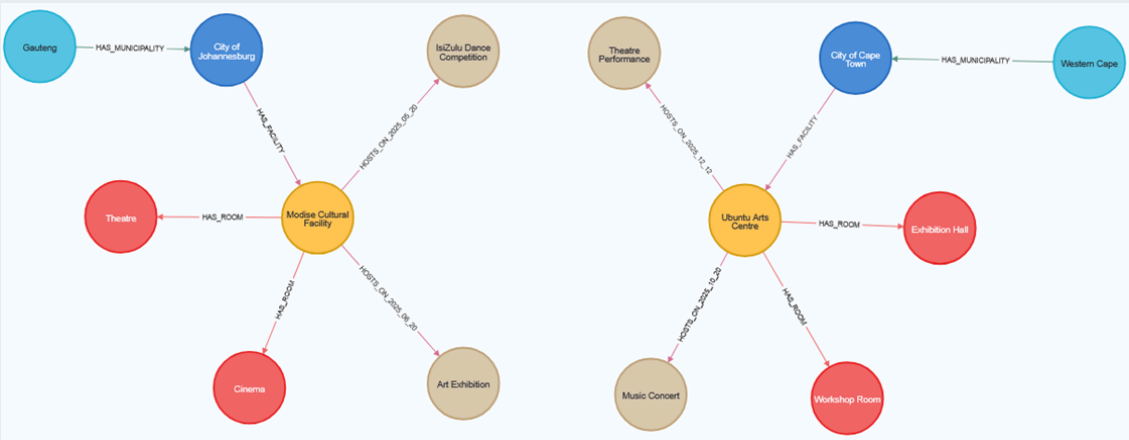
(d) Data nodes of facility and activity room as well as associated relationship. The output of the query results may be displayed in the format below:





(8 marks)

3.3 Using query, display all the created collections (nodes) and their relationships in a graph data model as shown in the figure below. The result output of the query for **all nodes and relationships in graphical** may be displayed in the format below:



(2 Marks)

3.4 Create a script to display the total capacity of facilities within each municipality. The result of your query may be displayed in the table format below:

	MunicipalityName	TotalCapacity
1	"City of Johannesburg"	500
2	"City of Cape Town"	300

(4 Marks)

3.5 Retrieve and display a list of all facilities along with their municipalities and provinces. The result of the query/script obtained from three (3) nodes (collections) may be displayed in any of the format below:

List of facilities along with their municipalities and provinces in text format:

FacilityName	MunicipalityName	ProvinceName
"Modise Cultural Facility"	"City of Johannesburg"	"Gauteng"
"Ubuntu Arts Centre"	"City of Cape Town"	"Western Cape"

List of facilities along with their municipalities and provinces in table format:

	FacilityName	MunicipalityName	ProvinceName
1	"Modise Cultural Facility"	"City of Johannesburg"	"Gauteng"
2	"Ubuntu Arts Centre"	"City of Cape Town"	"Western Cape"

(6 Marks)

3.6 Write a query to determine and display all activities along with the number of days until they occur from today. The result of the query may be displayed in any of the format below:

ActivityName	FacilityName	ActivityDate	DaysUntilActivity
"IsiZulu Dance Competition"	"Modise Cultural Facility"	"2025-05-20"	31
"Art Exhibition"	"Modise Cultural Facility"	"2025-06-20"	62
"Music Concert"	"Ubuntu Arts Centre"	"2025-10-20"	184
"Theatre Performance"	"Ubuntu Arts Centre"	"2025-12-12"	237

	ActivityName	FacilityName	ActivityDate	DaysUntilActivity
1	"IsiZulu Dance Competition"	"Modise Cultural Facility"	"2025-05-20"	31
2	"Art Exhibition"	"Modise Cultural Facility"	"2025-06-20"	62
3	"Music Concert"	"Ubuntu Arts Centre"	"2025-10-20"	184
4	"Theatre Performance"	"Ubuntu Arts Centre"	"2025-12-12"	237

(5 Marks)

3.7 Write a query to retrieve and display a list of municipalities with an average population of 4,000,000 (4 million) or more. The result of the query may be displayed in any of the format below:

Table

Text

Code

MunicipalityCode	MunicipalityName	AveragePopulation
"JHB"	"City of Johannesburg"	4949347
"CPT"	"City of Cape Town"	4005016

Table

Text

Code

	MunicipalityCode	MunicipalityName	AveragePopulation
1	"JHB"	"City of Johannesburg"	4949347
2	"CPT"	"City of Cape Town"	4005016

(5 Marks)

End of Question 3

8. Addendum

Good day,

Please check this page regularly for any update made on the project.

Thank you.